A First Course in Abstract Algebra

Section 17: Applications of G-Sets to Counting

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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17 Applications of G-Sets to Counting

Counting method. All these exercises request Burnside's formula:

$$\#\text{orbits} = \frac{1}{|G|}\sum_{g\in G}|X_g|.$$

Choose the labeled objects X and the allowed symmetry group G first. A coloring fixed by a permutation must be constant on each cycle of that permutation. With q available colors and c cycles, there are q^c fixed colorings when repetition is allowed. For three-dimensional solids we use rigid rotations; a square piece of cardboard can also be turned over, which induces the plane reflections of its edges.

Exercise 17.1: A cyclic action on eight points

The generator $(1,3,5,6)$ has order four, so the group contains its zeroth, first, second, and third powers. The identity fixes all 8 points. Each of the other three powers moves all four points $1,3,5,6$ and fixes $2,4,7,8$, so each has 4 fixed points. Burnside gives

$$\#\text{orbits} = \frac{8+4+4+4}{4} = \boxed{5}.$$

As a check, the orbits are $\{1,3,5,6\}$ and the four singleton fixed points.

Exercise 17.2: Two generators with disjoint supports

Let $a=(1,3)$ and $b=(2,4,7)$. They commute, have orders 2 and 3, and generate the six distinct elements a^ib^j with $i=0,1$ and $j=0,1,2$. For a point in $\{1,3\}$ to be fixed we need $i=0$; this contributes two fixed points for each such element. A point in $\{2,4,7\}$ is fixed exactly when $j=0$, contributing three. The points $5,6,8$ are always fixed. Thus, in the order e,a,b,b^2,ab,ab^2, the fixed counts are $8,6,5,5,3,3$. Consequently

$$\#\text{orbits} = \frac{8+6+5+5+3+3}{6} = \boxed{5}.$$

Exercise 17.3: Tetrahedral dice with four different labels

The rotation group of a regular tetrahedron has 12 elements: one can choose the face on the bottom in four ways and rotate it into three orientations. There are $4!=24$ labeled placements of the four different face labels. The identity fixes all 24. Every nonidentity rotation permutes at least two faces, so it cannot fix a placement with distinct labels. Burnside therefore gives

$$\frac{24+11\cdot 0}{12} = \boxed{2}$$

distinguishable dice. Reflections are not rigid rotations of a solid die.

Exercise 17.4: Cube faces with six different colors

There are $8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 20160$ assignments of six distinct colors to six specified faces. The rotation group has order 24. Only the identity can fix a coloring with all faces different, because any other rotation moves some face to another face of a different color. Burnside gives $20160/24 = \boxed{840}$ distinguishable blocks.

Exercise 17.5: Cube faces with arbitrary color repetition

Plan. Divide the 24 rotations into geometric types and count their cycles on the six faces. A fixed coloring is constant on each face cycle.

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Quarter-turn about face centers	6	$1,1,4$	3
Half-turn about face centers	3	$1,1,2,2$	4
Third-turn about opposite vertices	8	$3,3$	2
Half-turn about opposite edges	6	$2,2,2$	3

For instance a face-axis quarter-turn fixes the two axial faces and cycles the other four, giving three freely chosen colors. The entries in the second column sum to 24. With $q = 8$, Burnside gives

$$\frac{8^6 + 6 \cdot 8^3 + 3 \cdot 8^4 + 8 \cdot 8^2 + 6 \cdot 8^3}{24} = \frac{281088}{24} = \boxed{11712}.$$

Repeated colors are allowed independently on different cycles.

Exercise 17.6: Coloring the eight vertices of a cube

The same rotations now act on vertices, with different cycle counts:

Rotation type	Number	Vertex cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1,1,1$	8
Face-axis quarter-turn	6	$4,4$	2
Face-axis half-turn	3	$2,2,2,2$	4
Vertex-axis third-turn	8	$1,1,3,3$	4
Edge-axis half-turn	6	$2,2,2,2$	4

For example, a third-turn fixes its two axial vertices and cycles the other six in two triples. With four available colors, Burnside gives

$$\frac{4^8 + 6 \cdot 4^2 + (3+8+6)4^4}{24} = \frac{69984}{24} = \boxed{2916}.$$

Here a color need not appear; if it does, it may appear on any number of vertices. If one additionally requires all four colors to appear, which is a different restriction, replace each 4^c by the number of onto maps from its c cycles to four colors; the result would be 1718.

Exercise 17.7: Painting the edges of a square

The allowed group is D_4, of order 8. Turning over the cardboard makes reflections physically available as well as in-plane rotations.

(a) With no repeated color there are $6 \cdot 5 \cdot 4 \cdot 3 = 360$ labeled paintings. Only the identity fixes any of them. Hence Burnside gives $360/8 = \boxed{45}$.

(b) On the four edges, the identity has four cycles; two quarter-turns have one cycle each; the half-turn has two cycles; two reflections have two cycles; and two reflections have three cycles. Therefore

$$\frac{6^4 + 2 \cdot 6 + (1+2)6^2 + 2 \cdot 6^3}{8} = \frac{1848}{8} = \boxed{231}.$$

A reflection through opposite side midpoints fixes those two edges and swaps the other two, accounting for the three-cycle type.

Exercise 17.8: Resistors on the edges of a tetrahedron

Let X be the 2^6 assignments of two resistor types to the six edges. Use the 12 rigid rotations of the tetrahedral framework. The identity has six edge cycles. Each of the eight third-turns around a vertex and the opposite face center divides the edges into two triples, hence has two cycles. Each of the three half-turns around opposite edge midpoints fixes those two edges as sets and swaps the other four in two pairs, hence has four cycles. Burnside gives

$$\frac{2^6 + 8 \cdot 2^2 + 3 \cdot 2^4}{12} = \frac{144}{12} = \boxed{12}.$$

The objects are actual frameworks in space, so mirror reflection is not an additional rigid rotation. If mirror images were expressly identified, the larger symmetry group would instead give 11 classes.

Exercise 17.9: Coloring a prism with two square ends

Identify the rotation group.

The unique long axis must map to itself, so arbitrary cube rotations cannot be used. There are four rotations about the long axis, through $0, 90, 180, 270$ degrees, and four half-turns about transverse axes. Thus $|G| = 8$. On the six faces the cycle counts are as follows:

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Long-axis quarter-turn	2	$1,1,4$	3
Long-axis half-turn	1	$1,1,2,2$	4
Transverse half-turn through side centers	2	$1,1,2,2$	4
Transverse half-turn through side-edge midpoints	2	$2,2,2$	3

The transverse rotations interchange the two square ends. In the first transverse type, two lateral faces remain individually fixed; in the second type the four lateral faces are interchanged in pairs.

(a) With all six colors distinct there are $6! = 720$ labeled colorings, and only the identity fixes any. Burnside gives $720/8 = \boxed{90}$.

(b) With repetition, each cycle can have any of six colors. Hence

$$\frac{6^6 + 3 \cdot 6^4 + 4 \cdot 6^3}{8} = \frac{46656 + 3888 + 864}{8} = \frac{51408}{8} = \boxed{6426}.$$

Answer-key discrepancy. The scanned book's odd-answer key prints 6246 for part (b). The rotation table above gives 6426. Independently, enumerating all 46656 labeled face colorings and identifying them under the eight signed-coordinate rotation matrices preserving the long axis also gives 6426 orbits. Thus the printed answer has transposed digits.